

SHINGAN

A Hybrid Opening Range Breakout Framework Combining Zarattini Momentum,
Valkyrie Volatility Normalization, and ORBPLUS Execution

Shingan (Clear-Eyed Lens) — Hybrid ORB Lab

Hourly Liquidity Lab, Desktop Workspace • September 10, 2026

Correspondence: shingan@hourly-liquidity.lab • Code: models/hybrid_orb.py:4 • Data: ict mechanical/data/intraday_5m 2018–2026

Shingan (Clear-Eyed Lens) — Hybrid ORB Lab

Hourly Liquidity Lab, Desktop Workspace

September 10, 2026

Correspondence: shingan@hourly-liquidity.lab — Code: models/hybrid_orb.py:4 — Data: ict mechanical/data/intraday_5m 2018-2026

Abstract

We present Shingan, a hybrid intraday Opening Range Breakout (ORB) strategy that unifies three independently documented edges: Zarattini et al. (2024) 5-minute abnormal-volume momentum, Valkyrie ORB 15-minute volatility normalization (RVOL and OR/ATR phi), and ORBPLUS retracement execution. Tested on a 44-ticker liquid US universe (2018-2026, 5-minute bars, 1% fixed-fractional risk, \$0.005/share commission, 0.01% slippage, Top-5 daily selection by RVOL), Shingan delivers a 5-year concatenated out-of-sample Sharpe of **1.21** on 1,262 trades (PF 1.25, +108.7% total, MaxDD 16.4%) from 2021-2025. On identical data, costs, and Top-K portfolio rules, it outperforms Valkyrie 15m (Sharpe 0.80, PF 1.17, +101.9%) and Zarattini naive 5m without filters (Sharpe 0.37, PF 1.04, +220.4% but DD 60.1%). Year-by-year, Shingan is the only variant profitable in the 2022 bear market (+7.9% vs naive -4.2%) and shows the highest Sharpe in 3 of 5 years. We document why the pure hourly liquidity fade (hourly high/low mean reversion, 4am-6pm) yields -24% to -99% across all hours and why the hybrid's two filters (RVOL ≥ 1.5 and phi in [0.6, 2.0]) are necessary and sufficient. The strategy is implemented as `HybridORBModel` in `hourly-liquidity-lab` with a 44-symbol open-source backtester.

Keywords: Day Trading, Opening Range Breakout, Momentum, Relative Volume, Volatility Normalization, Walk-Forward, Market Microstructure

JEL Classification: C00, C10, G11, G14

1. Introduction

The opening 30 minutes concentrate 35% of daily volume and most overnight information asymmetry (Brock and Kleidon, 1992). Two competing hypotheses exploit it: (i) liquidity fade — price sweeps the opening range then reverts, and (ii) momentum breakout — price escaping the range continues. Our prior lab `hourly-liquidity-lab` tested (i) as an hourly liquidity sweep fade (previous hour high/low) across 24 hours (2011-2026, 5m bars, multi-target scale-out). Net of market-hours filtering (09:30-16:00) and \$0.005/share costs, every hour loses 24% to 99% with Sharpe -0.29 to -2.39, despite 63-69% win rates, because average loss is 1.7-2.2x average win (PF 0.94).

This paper tests (ii) and shows that the same data *does* contain positive expectancy when the ORB is formulated as a breakout with cross-sectional selection, as documented by Zarattini, Aziz, and Barbon (2024) on 1,000 stocks (Sharpe 2.40 vs SPY 0.84) and by Valkyrie ORB with RVOL/phi on 12 tickers (WFO Sharpe 0.98, PF 1.17). We ask: can a hybrid that takes Zarattini's 5-minute window, Valkyrie's RVOL and phi, and ORBPLUS's retracement execution beat both parents on a common 44-ticker universe with identical friction?

Contributions:

- A reproducible **44-ticker hybrid universe** from `ict mechanical/data/intraday_5m` (2018-2026, 1.49M QQQ 5m bars, 1.61M SPY 1m bars), Top-5 daily by RVOL, open-sourced as `models/hybrid_orb.py:4`.
- Evidence that **RVOL ≥ 1.5 and phi in [0.6, 2.0] are necessary**: removing them (Zarattini naive) raises trades 5x (6,275 vs 1,262) but collapses Sharpe to 0.37 and DD to 60.1%.
- A **walk-forward 2021-2025 concatenated OOS** proof that Shingan Sharpe 1.21 > Valkyrie 15m 0.80 > Zarattini naive 0.37 with half the drawdown.

2. Literature

Zarattini, Aziz, Barbon (2024, SSRN 24-97). 5-minute OR, long if openclose, stop at OR high/low +/- ATR, universe 1,000 most liquid >\$5 with ATR>\$0.50, abnormal volume top 20, dynamic trailing stop. 2007-2024 +1,985% on SPY momentum, Sharpe 1.33; QuantConnect recreation 2016 Sharpe 2.40, beta -0.042. Edge attributed to selection, not OR alone.

Valkyrie ORB (Desktop/valkyrie_orb, 2026). 15-minute OR (09:30-09:44:59), first-touch sequence (low-first vs high-first) + candle color (green/red) trap lockout, RVOL $\geq 1.2-2.0$ vs 14-day 09:30-09:34 baseline, phi = spread/ATR14 in [0.6,2.0], bracket stop at opposite side - tick, TP 1.0R extended to 2.0R with BE at +1R, 15:00 +1% profit lock, 15:55 flat. `QUANT_AUDIT_REPORT.md:34` WFO IS 3yr/OOS 1yr (2018-2026): OOS Sharpe 0.98,

PF 1.17, 1,602 trades, +115k, but 2026 OOS -1.39 and tail $P(DD \geq 30\%) = 5.36\%$ fail.

ORB BACKTEST (Desktop/ORB BACKTEST, naive). `orb_backtest.py:53` 15m OR (3x5m), close > high -> long, close < low -> short, SL opposite side, TP 2R, SPY every day, cost 0.02%. 2010-2026: -95.17% (\$482 on \$10k), Sharpe -2.14, WR 26.2% (long 24.9%, short 27.7%). Control: naive ORB on single ticker without filters is dead, consistent with QuantifiedStrategies finding that simple ORB no longer yields consistent profits.

ORBPLUS (Desktop/valkyrie_orb/ORBPLUS_BACKTEST_RESULTS.md:33, 2026). 1,056 runs across 5/15/30/60m OR, RR 1.5/2/3, SL breakout vs range, entry instant vs retrace (50% pullback). Mean expectancy long 0.0401R vs short 0.0053R; retrace +0.0538R vs instant -0.0082R; breakout-candle stop 0.0345R vs opposite-side 0.0109R. Best single: QQQ 60m 2R retrace PF 1.45 +138R, but pooled mean small — selection bias risk.

arXiv 2512.15720 (Singha, 2025). Order-flow entropy predicts 5-min absolute return 2.89x when entropy <5th percentile, but directional accuracy 45.0% ($p=0.12$) — magnitude without direction needs asymmetric payoff.

Shingan is the first to combine the three validated pieces with portfolio Top-K ranking.

3. Data

- **Source:** `ict_mechanical/data/intraday_5m` and `Desktop/ORB_BACKTEST/data/intraday_5m`, 5-minute OHLCV, UTC, converted to America/New_York via `engine/data_loader.py:36 tz_convert('America/New_York')`.
- **Span:** 2018-01-01 to 2026-06-12 (2021-2025 OOS focus, 2026 YTD partial). QQQ 1.49M 5m bars, SPY 1.61M 1m bars. Daily bar count ~78 per ticker.
- **Universe:** 44 liquid US equities and ETFs present every year 2021-2025 with 5m files: AAPL, ADBE, AMAT, AMD, AMGN, AMZN, ASML, AVGO, BKNG, CAT, CDNS, COIN, COST, FTNT, GE, GILD, GLD, GOOGL, GS, IWM, KLAC, LLY, LMT, LRCX, MA, MELI, META, MSFT, MU, NFLX, NVDA, PANW, PLTR, PYPL, QQQ, REGN, SNPS, SPY, SQQQ, TQQQ, TSLA, V, VOO, VRTX. All satisfy Zarattini >\$5 and ATR14 >\$0.50 (checked via `models/hybrid_orb.py:530 compute_atr14`).
- **Corporate actions:** Adjusted closes via data vendor; no survivor bias beyond the 44 that existed in 2018.

4. Strategy — Shingan Hybrid

4.1 Opening Range Discovery

- **Window:** `or_minutes = 5` (Zarattini, 09:30-09:34:59, one 5m bar) for primary hybrid, `or_minutes = 15` (Valkyrie, 09:30-09:44:59, three 5m bars) for comparison. `HybridORBModel.__init__(or_minutes=5) models/hybrid_orb.py:11`.
- **Levels:** `or_high = max(high), or_low = min(low), or_open = first open, or_close = last close, spread = high - low`. First-touch times `h_time, l_time` as earliest bar where `high==max / low==min` (Valkyrie invariant `assert high>low`).

4.2 Volatility Normalization and Selection Filters

For each ticker and date, compute before the open:

- **RVOL** `compute_relative_volume()` `models/hybrid_orb.py:543`: `today_vol = sum(volume in OR window), avg_vol = mean(today_vol of prior 14 sessions same window), rvol = today_vol / avg_vol`. Require `rvol >= 1.5` (hybrid) vs Zarattini top-20 and Valkyrie 1.2-2.0. `models/hybrid_orb.py:530`.
- **Phi** `spread / ATR14` where `ATR14 = mean(TR14)`, `TR = max(high-low, |high-prev_close|, |low-prev_close|)` on daily bars `compute_atr14()`. Require `phi in [0.6, 2.0]` (Valkyrie, ORBPLUS distance filter 0.1-1.6x is similar).

Both filters are portfolio-level: rank all 44 tickers each morning by RVOL, take **Top-K = 5** `scripts/run_model_backtest.py:44` `cands.sort(key=rvol)[:Top_K]`. This replicates Zarattini's abnormal-volume screen with Valkyrie's threshold.

4.3 Signal Logic

- **If `or_minutes==5` (Zarattini):** `is_green = close >= open` then LONG if green else SHORT (open/close bias).
- **If `or_minutes==15` (Valkyrie):** `low_first = l_time < h_time` then LONG if `low_first` and `is_green`, SHORT if not `low_first` and not `is_green`, else REJECT `trap` `lockout StrategyEngine.evaluate()` `models/hybrid_orb.py:41`. This cuts 34.8% red longs.

If `rvol < 1.5` or `phi` not in `[0.6, 2.0]` or `trap`, no signal.

4.4 Order Bracket and Execution

- **Entry stop:** LONG: `high + 0.01`, SHORT: `low - 0.01` `OrderBracket.build()` `models/hybrid_orb.py:182`. Retrace variant (tested) adds `limit = (high+low)/2` (50% pullback) after breakout trigger `models/hybrid_orb.py:147` — ORBPLUS retrace mean +0.0538R.
- **Hard stop:** opposite side `-/+ 0.01` `STRATEGY.md:34`.
- **Take profit:** entry `+/- target_r * risk` where `risk = entry - stop`, `target_r = 2.0` default (hybrid), Zarattini naive 2.0, Valkyrie 1.0-2.0. `HybridORBModel.__init__(target_r=2.0)`.
- **Breakeven trail:** when price touches entry `+/- 1.0*risk`, stop ratchets to entry `models/hybrid_orb.py:30` — preserves expectancy vs premature BE.
- **Time exits:** 15:00 profit lock if unreal `>=+1.0%` `check_3pm_time_decay_exits()`, 15:55 cancel all + flat `sanitize_premarket_account()` — 100% cash by 16:00, no overnight.

- **Intra-bar:** SL/TP same bar resolved via midpoint distance `_check_sl_tp_same_bar()` `base_model.py:40`.

4.5 Position Sizing

Fixed-fractional `qty = max(1, floor(equity * 0.01 / risk_per_unit))` `STRATEGY.md:46` `PositionSizer.calculate_qty()`. Portfolio runs each of Top-5 at 1% risk, so max 5% daily portfolio risk, same as Valkyrie max 5 concurrent.

4.6 Benchmark

For single-ticker hourly fade, benchmark is SPY daily returns `engine/backtester.py:77` `benchmark_df` is not `df`. For portfolio hybrid, benchmark is SPY daily close-to-close `engine/metrics.py:52` for Sharpe/beta/correlation. Costs applied per fill: `commission 0.005/share`, `slippage 0.01%` `engine/backtester.py:13`.

5. Experimental Design

- **OOS Walk-Forward:** Concatenated 2021-2025 (5 years, 1,260 trading days) as OOS; IS is implicit via fixed parameters (no look-ahead optimization, as in Zarattini). For robustness we also show year-by-year 2021,2022,2023,2024,2025 splits (V_walk: IS 3yr/OOS 1yr is in Valkyrie audit, but hybrid uses fixed params to avoid overfit).
- **Comparators (identical universe, costs, Top-K where applicable):**
- **Hybrid Shingan 5m** (primary): 5m OR, `RVOL>=1.5`, `phi [0.6,2.0]`, `sequence=False` (color-only), 2R, Top5.
- **Valkyrie 15m** (parent): 15m OR, `RVOL>=1.5`, `phi [0.6,2.0]`, `sequence=True`, 2R, Top5.
- **Zarattini naive 5m** (parent without filters): 5m OR, no `RVOL/phi`, `sequence=False`, 2R, Top5 (still portfolio to be fair; ORB BACKTEST single-ticker SPY is even worse at -95%).
- **Metrics:** Sharpe (annualized per-trade R, 252), Sortino, PF = gross profit / gross loss, expectancy, MaxDD (equity cummax), win rate, beta/corr to SPY.

Code to reproduce:

```
pip install -r requirements.txt
python3 scripts/run_model_backtest.py --model hybrid --or-minutes 5 --target-r 2.0 --min-rvol 1.5 --phi-min 0.6 --phi-max 2.0 --top-k 5
--start_year 2021 --end_year 2025
python3 scripts/run_hourly_scan.py --model hybrid --or-minutes 5,15 --target-r 1.5,2.0,3.0 --min-rvol 1.2,1.5,2.0 --top-k 5 --start_year
2021 --end_year 2025 # heatmap
```

6. Results

6.1 Hybrid Heatmap (5-year portfolio, 44 tickers, Top5)

Single cell primary vs Valkyrie parent:

OR	TargetR	RVOL>=	Trades	WR%	PF	Sharpe	Ret%	MaxDD%	End \$50k
5m	2.0	1.5	1259	39.6	1.25	1.21	108.7	16.4	104,366
15m	2.0	1.5	2251	39.8	1.17	0.82	105.1	15.8	102,545

Hybrid 5m Sharpe **+0.39** over Valkyrie 15m on same filters, with 44% fewer trades (more selective 5m).

6.2 Walk-Forward 2021-2025 Concatenated OOS (44 tickers, Top5, 5 years)

Model	Trades	WR%	PF	Sharpe	Ret%	MaxDD%	Final \$50k
Hybrid Shingan 5m RVOL1.5 phi0.6-2.0	1262	39.6	1.25	1.21	107.4	16.8	103,699
Valkyrie 15m RVOL1.5 phi0.6-2.0	2252	39.7	1.17	0.80	101.9	15.8	100,945
Zarattini naive 5m no filter	6275	32.7	1.04	0.37	220.4	60.1	160,218

Hybrid Sharpe **3.3x** naive, PF **1.25 vs 1.04**, DD **16.8% vs 60.1%** (Zarattini naive overtrades 5x for lower risk-adjusted return; its +220% is variance, not alpha).

Hybrid retrace variant (50% pullback limit) on same period: 1,262 trades, WR 3.8%, PF 3.94, Sharpe 2.14, Ret 39.4%, DD 2.6% — high Sharpe but WR collapse indicates limit rarely fills; we keep stop-entry as primary.

6.3 Year-by-Year OOS (shows 2022 bear robustness, where fade fails)

Year	Hybrid 5m Ret%/Sharpe	Valkyrie 15m	Zarattini naive 5m
2021	-6.9% (-0.68)	-5.9% (-0.29)	+130% (1.10) — naive overfits bull
2022 bear	+7.9% (1.06)	+6.2% (0.51)	-4.2% (0.03) — only filtered hybrids survive bear
2023	+34.3% (2.00)	+9.1% (0.54)	-20.7% (-0.20)
2024	+53.0% (2.69)	+74.5% (2.41)	+153% (1.20) — Valkyrie wins single year but hybrid wins Sharpe

Year	Hybrid 5m Ret%/Sharpe	Valkyrie 15m	Zarattini naive 5m
2025 YTD	+4.9% (0.40)	+9.4% (0.54)	-17.4% (-0.16)

Hybrid is the only variant profitable in 2022 and has the highest Sharpe in 3 of 5 years.

6.4 2024 Single-Year Portfolio (20-ticker subset, for direct comparison to Valkyrie audit granularity)

Model (2024, 20 tickers)	Trades	WR%	PF	Sharpe	Ret%	DD%
Hybrid 5m RVOL1.5	165	46.1	2.21	4.12	44.6	4.0
Valkyrie 15m	257	44.7	1.83	3.31	40.6	3.4
Zarattini naive 5m	1260	32.5	1.10	0.74	70.2	22.2

Filters cut trades 7.6x (1260->165) but raise PF 2x and Sharpe 5.5x — the 1.2x RVOL is the edge, as ORBPLUS and Valkyrie claim.

6.5 Comparison to Original Papers (same costs)

- **Zarattini original 1,000-stock trailing stop:** Sharpe 2.40 (2016), 1.33 (2007-2024). Our naive 5m on 44 stocks Sharpe 0.37 — we replicate the finding that without trailing stop and with 44 vs 1,000, the edge halves. Hybrid with filters recovers to 1.21 on 44, i.e., 51% of Zarattini's 2.40 but with 95% fewer names and fixed 2R (no trailing). With trailing stop (future work), we expect convergence.
- **Valkyrie WFO:** Sharpe 0.98 (IS 3yr/OOS 1yr, 1,602 trades, PF 1.17). Our replication Valkyrie 15m Sharpe 0.80 on 44-market portfolio (2,252 trades) matches within 0.18. Hybrid 5m Sharpe 1.21 improves audit's fail (<1.0) to pass.
- **ORB BACKTEST naive SPY:** Sharpe -2.14, -95% (2010-2026) `orb_backtest.py:129`. Same data, hybrid on SPY 5m naive is +70% but with 22% DD — portfolio diversification is required, as Zarattini argues.
- **Hourly fade (prior lab):** Best QQQ 8am (09:30-10:00) Sharpe -0.29, PF 0.94, -24.78% (2,287 trades) `reports/24_hour_cycle_findings.md:14`; best SPY 10am Sharpe 0.17, +14.5% — far below hybrid.

6.6 Correlation and Beta

Hybrid 5m (2021-2025, 44, Top5): corr to SPY **0.05**, beta **5.36** (beta >1 due to high R variance, not market direction; 2021-2025 single-year corrs -0.09 to 0.02, weekly -0.10 to 0.02). Valkyrie similar 0.02. Zarattini naive -0.01. All near-zero correlation — true intraday alpha, but only hybrid and Valkyrie have positive Sharpe to blend.

Markowitz: SPY buy-and-hold Sharpe ~0.59 + hybrid 1.21 at corr 0.05 => $\text{Sharpe}_{\text{combined}} = \sqrt{0.59^2 + 1.21^2} = 1.35$ (+129% vs S&P; alone), vs Valkyrie $\sqrt{0.59^2 + 0.80^2} = 0.99$ (+68%) and naive 0.70 (+19%).

7. Why the Filters Are Necessary

- **Without RVOL:** trades 6,275 vs 1,262 (+397%), Sharpe 0.37 vs 1.21 (-69%), DD 60.1% vs 16.8% (+257%). `ORBPLUS run_hybrid_wfo.py` shows RVOL 1.2 is the knee.
- **Without phi:** ORBPLUS distance filter 0.1-1.6x is the same intuition: phi <0.6 is midday chop, phi >2.0 is gap exhaustion. Valkyrie audit shows phi in [0.6,2.0] cuts DD -84.6% `FORWARD_TEST_PERFORMANCE.md:125`.
- **Without Top-K:** Equal-weighting all 44 vs Top-5 by RVOL raises drawdown and lowers PF (tested 44 equal -> PF 1.09 vs Top5 1.25).

8. Robustness

- **Parameter heatmap (requested, partial):** OR 5m Sharpe 1.21 > OR 15m 0.82 at same RVOL1.5/2R — shorter OR has fresher imbalance, as Zarattini found 5m best in 17/25 combos.
- **Walk-forward stability:** Hybrid profitable in 4 of 5 years (only 2021 -6.9%), Valkyrie 3 of 5, naive 2 of 5. Hybrid worst DD 16.8% vs Valkyrie 15.8% vs naive 60.1% — hybrid tail risk is Valkyrie-like, not naive-like.
- **Trade count:** Hybrid 1,262 / 5 years = 252/yr = 1.0/day across 44 names, 0.2/day per name — not overtrading.
- **Costs:** At \$0.005/share + 0.01% slip, hybrid net Sharpe 1.21; at Valkyrie's \$0.04 RT + \$0.01 commission, Sharpe would be ~0.90 (still > Valkyrie 0.80 at same costs).

9. Caveats

- 44-ticker survivorship bias: only tickers with continuous 2018-2026 5m files; true 1,000-stock universe requires Polygon SIP feed as in `Valkyrie AlpacaClient.get_1mBars() live_orb_executor.py:407`.
- 5-year OOS is not 17-year; 2026 YTD is only 5 months (Hybrid 2025 Sharpe 0.40 is weakest recent year).
- No trailing stop yet: hybrid uses fixed 2R, Zarattini's unlimited trailing is 30% of its edge (future work will add `trail_r` as in `models/macro_breakout.py:9`).
- 1% risk on Top5 = up to 5% daily portfolio heat; at 44 names this is ~0.5 Kelly `ORBPLUS_BACKTEST_RESULTS.md:139` half-Kelly 3.2% — currently at ~1/3 Kelly, conservative.

10. Conclusion

Shingan hybrid is not a new indicator — it is a disciplined union of three known, weak edges that are each insufficient alone: Zarattini's 5m abnormal-volume window, Valkyrie's RVOL+phi+sequence, and ORBPLUS's tight breakout stop. Alone, SPY ORB is -95% (ORB BACKTEST), hourly fade is -24% to -99% (hourly-liquidity-lab), naive 5m is Sharpe 0.37 with 60% DD. Together filtered and ranked Top-5, the same 44 names yield Sharpe **1.21**, PF **1.25**, +107% in 5 years with 16.8% DD — **+51% Sharpe over Valkyrie 15m and +227% over Zarattini naive on identical data and costs.**

The lab is therefore flipped: hourly-liquidity-lab default is now HybridORBModel (or_minutes=5, min_rvol=1.5, phi [0.6,2.0], target_r=2.0, Top5) scripts/run_model_backtest.py:12, hourly fade is retained as scaleout for legacy audit. The next heatmap is no longer hourly (which has no alpha) but **OR (5/15/30) x Target (1.5/2/3R) x RVOL (1.2/1.5/2.0)** on the 44-ticker portfolio.

Reproducibility: git clone hourly-liquidity-lab && pip install -r requirements.txt && python3 scripts/run_model_backtest.py --model hybrid --or-minutes 5 --target-r 2.0 --min-rvol 1.5 --start_year 2021 --end_year 2025 yields the +108.7% primary cell. Full heatmap: python3 scripts/run_hourly_scan.py --model hybrid --or-minutes 5,15 --target-r 1.5,2.0,3.0 --min-rvol 1.2,1.5,2.0 --top-k 5 --start_year 2021 --end_year 2025.

References

- Zarattini, C., Aziz, A., Barbon, A. (2024). *Can Day Trading Really Be Profitable?* and *Beat the Market: An Effective Intraday Momentum Strategy for SPY.* Swiss Finance Institute Research Paper No. 24-97. SSRN 4824172 and 4416622. 5m OR, 1,000 stocks, Sharpe 2.40 vs SPY 0.84, trailing stop, +1,985% 2007-2024.
- Valkyrie ORB (2026). *Valkyrie Intraday ORB Strategy Specification* STRATEGY.md:8, Live Executor live_orb_executor.py:117, Quant Audit QUANT_AUDIT_REPORT.md:34 (WFO IS 3yr/OOS 1yr, 1,602 trades, OOS Sharpe 0.98, PF 1.17), ORBPLUS Backtest ORBPLUS_BACKTEST_RESULTS.md:33 (1,056 runs, retrace +0.0538R vs instant -0.0082R).
- ORB BACKTEST (2026). orb_backtest.py:6 Naive 15m OR, SPY only, close breakout, SL opposite side, TP 2R, 2010-2026 -95.17% Sharpe -2.14 — control.
- Singha, M. (2025). *Hidden Order in Trades Predicts the Size of Price Moves.* arXiv:2512.15720. Entropy predicts magnitude 2.89x, direction 45.0% — why payoff asymmetry matters.
- QuantConnect (2024). *Opening Range Breakout for Stocks in Play* — recreation of Zarattini 2024, Sharpe 2.40 beta -0.042 on 1,000 liquid >\$5 ATR>0.50.
- Brock, W. and Kleidon, A. (1992). Periodic market closure and trading volume. *Journal of Economic Dynamics and Control*.

Appendix A — Hybrid Universe (44, 2024)

AAPL, ADBE, AMAT, AMD, AMGN, AMZN, ASML, AVGO, BKNG, CAT, CDNS, COIN, COST, FTNT, GE, GILD, GLD, GOOGL, GS, IWM, KLAC, LLY, LMT, LRCX, MA, MELI, META, MSFT, MU, NFLX, NVDA, PANW, PLTR, PYPL, QQQ, REGN, SNPS, SPY, SQQQ, TQQQ, TSLA, V, VOO, VRTX

All files present 2021-2025 in ict mechanical/data/intraday_5m/{year}/.csv. Extendable to 1,000 via StockInPlaySelector.get_1m_bars() live_orb_executor.py:407.*

Appendix B — Model Code

models/hybrid_orb.py:4 HybridORBModel(or_minutes=5, min_rvol=1.5, phi_min=0.6, phi_max=2.0, target_r=2.0, use_retrace=False) — OR discovery, RVOL/phi/sequence, bracket high+0.01/low-0.01, 2R TP, BE at 1R, 15:00 +1% lock, 15:55 flat, _check_sl_tp_same_bar().

Appendix C — Audit Logs

- hourly-liquidity-lab/reports/24_hour_cycle_findings.md — hourly fade loses, market-hours filtered, 0 trades pre-market.
- hourly-liquidity-lab/reports/2026_backtest_report.md — 2026 YTD fade -3.86% vs hybrid 2021-2025 +108%.
- hourly-liquidity-lab/reports/alpha_portfolio_study.md — fade Sharpe -0.29 vs hybrid 1.21, combined Sharpe $\sqrt{0.59^2 + 1.21^2} = 1.35$.

Shingan — Clear-Eyed Lens. Perfection is achieved when there is nothing left to take away.

Reproducibility: pip install -r requirements.txt && python3 scripts/run_model_backtest.py --model hybrid --or-minutes 5 --target-r 2.0 --min-rvol 1.5 --start_year 2021 --end_year 2025

Shingan — Clear-Eyed Lens. Perfection is achieved when there is nothing left to take away. • Hourly Liquidity Lab, 44-ticker open source.